Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Second Semester M.Tech Degree (S, FE) Examination January 2024 (2022 Scheme)

Discipline: COMPUTER SCIENCE AND ENGINEERING

Course Code & Name: 222TCS002 DEEP LEARNING

Max. Marks: 60 Duration: 2.5 Hours

PART A

Answer all questions. Each question carries 5 marks Marks

1		Explain how activation functions contribute to the operation of a neural network, and analyze their role in introducing non-linearities to improve the network's capacity to comprehend intricate information	(5)
2		Illustrate convolution and pooling operation with an example.	(5)
3		Contrast feed-**forward** network and recurrent neural network	(5)
4	a.	Explain unidirectional and bidirectional contextual models with the help of an example	(3)
	b.	Classify whether **BERT** is a contextual model or a context-free model.	(2)
5		Evaluate the rationale behind the purpose of the design of Transformer architecture, its advantages, and disadvantages with other neural network architectures like RNNs or CNNs.	(5)

PART B

Answer any 5 questions. Each question carries 7 marks

6	a.	Apply your understanding of optimization algorithms in machine learning by distinguishing between Batch Gradient Descent and Stochastic Gradient Descent.	(7)
	b.	Describe how these two approaches differ in optimizing the parameters of a model and highlight the practical implications of choosing one over the other in various scenarios.	
7	a.	Explain sparse encoders	(3)
	b.	In AlexNet, the input image is of size 227x227x3. The first convolutional layer has 96 kernels of size 11x11x3. The stride is 4 and the padding is 0. Determine the size of the output image right after the first bank of convolutional layers.	(2)

c. The MaxPool layer after the bank of convolution filters has a pool size of 3 and stride of 2. The image at this stage is of size 55x55x96. Calculate the output image after the MaxPool layer is of size (2)

8 Explain $\mathbf{L^1}$, $\mathbf{L^2}$ Regularization, Dropout, and Data augmentation (7)

9 Explain the functioning of Recurrent Neural Networks (RNNs) and discuss the utilization of backpropagation through time (BPTT) in recurrent networks. (7)

10 Demonstrate your understanding of the sequence-to-sequence architecture by explaining its encoding and decoding processes using a relevant example. Additionally, explore how this architecture functions and highlight its practical applications. (7)

11 a. Justify the necessity of employing Word embedding. (L4) (2)

b. Illustrate any one of the word2vec techniques of word embedding. (5)

12 a. Contrast BERT Base and BERT Large using diagrams. (3.5)

b. Elaborate on how the Bahdanau attention mechanism operates and discuss its essential components. (3.5)

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